

Contact-line dynamics in the kinematic-match model of drop rebound: finite contact angle and linear contact-line friction

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The kinematic-match (KM) reduced-order model of Gabbard *et al.* (2025) (hereafter GA) predicts drop rebound from a *perfectly non-wetting* substrate with no fitting parameters. Its non-wetting character is encoded in a single boundary condition: the free surface departs the substrate tangentially at the contact line. We relax this assumption from first principles, admitting substrates of finite equilibrium contact angle θ_e and a finite contact-line mobility. We frame the problem in a metriplectic-style (GENERIC-flavoured) form: the reversible capillary–inertial mode oscillator of GA, a unilateral non-penetration constraint whose Lagrange multiplier is the contact pressure, and a new *irreversible* dissipation localised at the moving contact line. Two results follow. First, the tangency condition is replaced by a dynamic contact-angle condition; in the quasi-static (perfectly mobile) limit this reduces to imposing the apparent contact angle $\theta_d = \theta_e$, and for $\theta_e \rightarrow \pi$ it recovers GA’s tangency exactly. Second, the uncompensated Young force is radial and cancels by axisymmetry, so *the centre-of-mass momentum equation of GA is unchanged*; the entire wetting effect enters through (i) the contact-line boundary condition and (ii) a contact-line dissipation channel governed by a linear (Blake/de Gennes) friction law $\xi \dot{r}_c = \sigma(\cos \theta_e - \cos \theta_d)$. The extended model has exactly one new dynamical degree of freedom (the contact-line radius r_c) and two physical parameters, the measurable θ_e and a single friction coefficient ξ ; it reduces to GA in the limit ($\theta_e \rightarrow \pi$, $\xi \rightarrow 0$). We give the discrete linear system, showing the extension changes only the contact-set *selection criterion* (quasi-static) plus a rate constraint (dynamic), preserving the linearity that underlies GA’s numerical robustness.

1. Introduction

A drop striking a solid or liquid surface may rebound without ever coalescing, a phenomenon central to superhydrophobic surface design, spray cooling, inkjet printing and self-cleaning materials (Richard *et al.* 2002; Snoeijer & Andreotti 2013). When rebound is mediated by a thin gas film the impact is effectively *non-wetting*: no three-phase contact line forms, and the drop’s contact time and coefficient of restitution are set by the interplay of inertia, capillarity and viscosity (Richard *et al.* 2002; Okumura *et al.* 2003; Moláček & Bush 2013). Reduced-order “kinematic-match” (KM) models exploit this by evolving the drop’s spherical-harmonic shape modes subject to a unilateral, tangential contact with the substrate, and reproduce experiment across a wide range of Weber number at a small fraction of the cost of direct numerical simulation (Galeano-Rios *et al.* 2017; Alventosa *et al.* 2023; Gabbard *et al.* 2025).

The defining assumption of these models is *perfect* non-wetting: the free surface meets

the substrate tangentially, so the equilibrium contact angle is $\theta_e = \pi$, there is no adhesion, and the contact line carries no localised dissipation. Real substrates — including the hydrophobic coatings used to suppress, but not eliminate, wetting — depart from this idealisation. A finite equilibrium contact angle introduces an uncompensated Young force at the contact line (de Gennes 1985), its motion dissipates energy through microscopic processes captured at leading order by a linear (molecular-kinetic / hydrodynamic) friction law (Blake 2006; Cox 1986; Voinov 1976; Snoeijer & Andreotti 2013), and the advancing and receding branches generically differ, producing contact-angle hysteresis (Joanny & de Gennes 1984). None of these ingredients is present in the KM framework.

Here we extend the KM model to substrates of finite wettability from first principles, retaining the linear structure responsible for its numerical robustness. We organise the dynamics in a metriplectic (GENERIC-flavoured) form (Grmela & Öttinger 1997): the reversible capillary-inertial mode oscillator of Gabbard *et al.* (2025) (hereafter GA), a unilateral non-penetration constraint whose Lagrange multiplier is the contact pressure, and an irreversible dissipation localised at the contact line. The tangency condition is replaced by a dynamic contact-angle condition closed by a Blake/de Gennes mobility law; the perfectly non-wetting model is recovered as the limit $\theta_e \rightarrow \pi$ with infinite mobility. We show that the extension is a change of *constraint* rather than an added body force, so the centre-of-mass equation and the discrete linear system of GA are structurally preserved.

2. The perfectly non-wetting model

We adopt the notation and non-dimensionalisation of GA Sec. 3.1: lengths by the undeformed radius R , mass by ρR^3 , time by the inertio-capillary time $t_\sigma = (\rho R^3 / \sigma)^{1/2}$, so that $\sigma \equiv 1$ in dimensionless units and gravity and viscosity are carried by $Bo = \rho g R^2 / \sigma$ and $Oh = \mu / \sqrt{\rho \sigma R}$. The axisymmetric interface is

$$\zeta(\theta, t) = 1 + \eta(\theta, t), \quad \eta = \sum_{l=2}^{\infty} \mathcal{A}_l(t) P_l(\cos \theta), \quad (2.1)$$

with P_l the Legendre polynomials and $\theta \in [0, \pi]$ the polar angle measured from the downward vertical, so that $\theta = 0$ is the pole facing the substrate. The contact pressure is $p(\theta, t) = \sum_{l \geq 0} \mathcal{B}_l(t) P_l(\cos \theta)$. With $\mathcal{U} = \dot{\mathcal{A}}$ the modal surface velocity, h the centre-of-mass (COM) height above the substrate and $v = \dot{h}$, the linearised dynamics of GA are

$$\dot{\mathcal{A}}_l = \mathcal{U}_l, \quad (2.2)$$

$$\dot{\mathcal{U}}_l = -l(l+2)(l-1) \mathcal{A}_l - 2(2l+1)(l-1) Oh \mathcal{U}_l - l \mathcal{B}_l, \quad (2.3)$$

$$\dot{h} = v, \quad (2.4)$$

$$\dot{v} = -Bo + \mathcal{B}_1. \quad (2.5)$$

Equations (2.2)–(2.5) hold for $l = 2, \dots, \infty$. The COM force (2.5) is the net vertical reaction of the contact pressure (its $l = 1$ moment) plus gravity; we return to this in §5.

Cartesian geometry. A surface point at polar angle θ sits at cylindrical radius and height

$$x(\theta) = \zeta(\theta) \sin \theta, \quad z(\theta) = h - \zeta(\theta) \cos \theta, \quad (2.6)$$

so the pole $\theta = 0$ is at height $z = h - \zeta(0)$ and the substrate is $z = 0$. The signed gap to the substrate is $z(\theta)$; contact requires $z = 0$ and impenetrability requires $z \geq 0$.

The kinematic match. Where the substrate presses the drop a contact cap $\theta \leq \theta_c$ forms on $z = 0$:

$$\sum_{l \geq 2} \mathcal{A}_l P_l(\cos \theta) = \frac{h}{\cos \theta} - 1, \quad \theta \leq \theta_c, \quad (2.7)$$

$$\sum_{l \geq 2} \mathcal{A}_l P_l(\cos \theta) < \frac{h}{\cos \theta} - 1, \quad \theta > \theta_c, \quad (2.8)$$

$$\sum_{l \geq 0} \mathcal{B}_l P_l(\cos \theta) = 0, \quad \theta > \theta_c, \quad (2.9)$$

where (2.8) is the strict form of $z > 0$ off contact (equivalently $\zeta \cos \theta < h$).[†] Finally GA impose *tangential departure* at the contact line,

$$\left. \frac{\partial}{\partial \theta} [\zeta(\theta) \cos \theta] \right|_{\theta=\theta_c} = 0, \quad (2.10)$$

which, since $\partial_\theta(\zeta \cos \theta) = -\partial_\theta z$, states $\partial_\theta z|_{\theta_c} = 0$: the gap has a smooth (tangential) zero. This is the sole encoding of perfect non-wetting; §3 replaces it.

3. The apparent contact angle

Define the *apparent (dynamic) contact angle* θ_d as the angle, measured through the liquid, between the substrate and the liquid–gas interface at the contact line. Using (2.6), the interface tangent in the direction of increasing θ (i.e. away from the wetted pole, up the free surface) is $\mathbf{T} = (x_\theta, z_\theta)$ with, writing $\zeta_\theta \equiv \partial_\theta \zeta = -\sin \theta \sum_{l \geq 2} \mathcal{A}_l P'_l(\cos \theta)$,

$$x_\theta = \zeta_\theta \sin \theta + \zeta \cos \theta, \quad z_\theta = \zeta \sin \theta - \zeta_\theta \cos \theta. \quad (3.1)$$

Let $\alpha = \text{atan2}(z_\theta, x_\theta)$ be the inclination of $\mathbf{T}$ above the substrate ($+x$). The liquid occupies the interior (axis) side, so the contact angle through the liquid is the supplement,

$$\boxed{\theta_d = \pi - \alpha = \pi - \text{atan2}(\zeta \sin \theta_c - \zeta_\theta \cos \theta_c, \zeta_\theta \sin \theta_c + \zeta \cos \theta_c).} \quad (3.2)$$

LEMMA 1 (UNDEFORMED REDUCTION). *For an undeformed sphere ($\mathcal{A}_l \equiv 0$, so $\zeta = 1$, $\zeta_\theta = 0$), $\theta_d = \pi - \theta_c$. In particular $\theta_c \rightarrow 0$ (point contact) gives $\theta_d \rightarrow \pi$ and $\theta_c \rightarrow \pi/2$ (hemisphere) gives $\theta_d \rightarrow \pi/2$.*

Proof. With $\zeta = 1, \zeta_\theta = 0$, (3.1) gives $(x_\theta, z_\theta) = (\cos \theta_c, \sin \theta_c)$, so $\alpha = \theta_c$ and $\theta_d = \pi - \theta_c$. $\square$

PROPOSITION 1 (TANGENCY IS THE $\theta_d = \pi$ CASE). *Assume the contact line lies on the ascending flank of the profile, $x_\theta(\theta_c) > 0$ (the cylindrical radius increases outward there). Then the GA tangency condition (2.10) holds iff $\theta_d = \pi$.*

Proof. $\partial_\theta z = z_\theta = \zeta \sin \theta_c - \zeta_\theta \cos \theta_c$ is the first argument of atan2 in (3.2). Under the hypothesis $x_\theta > 0$ the second argument is positive, so $z_\theta = 0 \iff \alpha = 0 \iff \theta_d = \pi$. By (2.10), $\partial_\theta z|_{\theta_c} = 0$, hence $\theta_d = \pi$, and conversely. $\square$

[†] The parent paper GA writes the non-contact inequality as $\zeta \cos \theta > h$ (their eqns for (2.8) and the impenetrability predicate e_q^1); evaluated at the drop apex $\theta = \pi$, where $z = h + \zeta > 0$ unambiguously, that form reads $-\zeta > h$, which is false. The correct sense is $\zeta \cos \theta < h$ as used here; GA's printed inequality is a sign typo (the two formulations agree, as they must, on the contact *equality* $\zeta \cos \theta = h$).

REMARK 1 (VALIDITY OF THE $x_\theta > 0$ HYPOTHESIS). $x_\theta = \partial_\theta(\zeta \sin \theta)$ is the slope of the cylindrical radius and vanishes at the profile's widest point: for the undeformed sphere $x_\theta = \cos \theta_c$, which is positive for $\theta_c < \pi/2$, zero at $\theta_c = \pi/2$, and negative beyond. The hypothesis therefore holds throughout the low- We , small-contact regime ($\theta_c < \pi/2$) that is our target, but can fail near maximal spreading or for pancake/overhanging shapes, where $\alpha = \text{atan2}(z_\theta, x_\theta)$ becomes ill-conditioned. The discrete residual of §8 is deliberately written in a form (8.1) that avoids dividing by x_θ and so remains well defined there.

Thus GA is exactly the special case $\theta_d \equiv \pi$. A substrate of finite wettability instead selects a finite apparent angle, determined dynamically below.

4. Energy structure and the wetting energy

Organise the model in a metriplectic-style (GENERIC-flavoured) form: reversible kinetic+surface energy, a unilateral constraint, and irreversible dissipation. We do not verify the formal GENERIC degeneracy conditions; the structure is used as an organising principle and is *backed* by the closed energy budget of §7. The total energy is

$$E = \underbrace{\frac{1}{2} \int_{\text{drop}} |\mathbf{u}|^2 dV}_{E_{\text{kin}}} + \underbrace{Bo}_{E_{\text{grav}}} h + \underbrace{A_{lg}}_{\text{liquid-gas}} + \underbrace{(\gamma_{sl} - \gamma_{sv}) A_w}_{\text{wetting}}, \quad (4.1)$$

in dimensionless units ($\sigma = 1$; γ_{sl}, γ_{sv} the solid-liquid and solid-vapour tensions in units of σ). Young's relation $\gamma_{sv} - \gamma_{sl} = \cos \theta_e$ turns the wetting term into

$$E_{\text{wet}} = -\cos \theta_e A_w = -\pi \cos \theta_e r_c^2, \quad r_c \equiv x(\theta_c) = \zeta(\theta_c) \sin \theta_c, \quad (4.2)$$

r_c being the contact-line radius and $A_w = \pi r_c^2$ the wetted area to leading order. The liquid-gas area, expanded to quadratic order in the $\mathcal{A}_l$, reproduces the Lamb capillary spectrum and hence the restoring term $-l(l+2)(l-1)\mathcal{A}_l$ already present in (2.3) (Lamb 1932); no new bulk term arises. The genuinely new ingredient is (4.2).

PROPOSITION 2 (REDUCTION OF THE WETTING ENERGY). *As $\theta_e \rightarrow \pi$, $\cos \theta_e \rightarrow -1$ and $E_{\text{wet}} \rightarrow \pi r_c^2$; its variations drive the contact line only through (6.4) below, whose driving force $\propto (\cos \theta_e - \cos \theta_d)$ vanishes when additionally $\theta_d \rightarrow \pi$ (Proposition 1). Hence at $(\theta_e \rightarrow \pi)$ with a tangential contact line the wetting contribution is dynamically inert, consistent with GA.*

5. The centre-of-mass equation retains the GA form

A central and non-obvious claim: the COM equation keeps the GA form $\dot{v} = -Bo + \mathcal{B}_1$ even for finite wettability — not because no capillary force acts at the contact line, but because that force is reacted by the constraint pressure and thereby absorbed into the pressure moment $\mathcal{B}_1$.

PROPOSITION 3 (COM EQUATION FORM IS PRESERVED). *The vertical force balance of the drop's centre of mass retains the GA form $\dot{v} = -Bo + \mathcal{B}_1$, where $\mathcal{B}_1$ is the $l = 1$ moment of the contact pressure. In particular the contact-line capillary reaction does not appear as a separate term.*

Proof. Take the drop (liquid plus its bounding liquid-gas interface) as the system. The external agents are gravity, the uniform vapour pressure (absorbed into the reference), and the solid. At the contact line the liquid-gas tension exerts a line force of magnitude

σ per unit length directed along the interface tangent; its *radial* (in-plane) component is the uncompensated Young force $\sigma(\cos\theta_e - \cos\theta_d)$ per length, which is horizontal and cancels by axisymmetry, while its *vertical* component is $\sigma\sin\theta_d$ per length, a net upward line load

$$F_{cl}^z = 2\pi r_c \sigma \sin\theta_d, \quad (5.1)$$

nonzero for any $\theta_d \neq \pi$. This vertical pull is not balanced by γ_{sl}, γ_{sv} (which are horizontal); it is balanced by the substrate *normal reaction*, i.e. by the contact pressure p , which is the Lagrange multiplier enforcing $z = 0$ on the wetted cap. In the reduced model the net vertical pressure reaction is exactly the $l = 1$ moment $\mathcal{B}_1$ (same projection as in GA), so (5.1) is subsumed into $\mathcal{B}_1$ rather than added separately, and $\dot{v} = -Bo + \mathcal{B}_1$ holds unchanged. $\square$

Consistency with GA. For $\theta_d = \pi$ (tangency) $\sin\theta_d = 0$ and $F_{cl}^z = 0$: GA's smooth-cap pressure carries no edge load, as expected. For finite θ_d the reaction includes a near-concentrated line load at θ_c .

REMARK 2 (NUMERICAL EDGE-LOAD CAVEAT). *Because the finite-angle reaction (5.1) concentrates near θ_c , the truncated degree- L Legendre pressure must represent a near-singular edge load. GA never faced this ($\sin\theta_d \equiv 0$). Expect Gibbs-type oscillation and $O(1/L)$ error in $\mathcal{B}_1$ for strongly non-tangential angles; the effect is $O(\sin\theta_d)$ and hence small in the near-non-wetting regime ($\theta_e \rightarrow \pi$) that is our experimental target. The energy-budget closure test (§7) is the quantitative bound on this error and should be monitored as θ_e departs from π .*

Remark (mechanism). The wetting energy (4.2) changes the dynamics *indirectly*: it alters the contact-line boundary condition (§6), which changes the admissible shape and hence the pressure multipliers $\mathcal{B}_l$ in (2.3) and (2.5). The extension is thus a change of *constraint*, not an added body force — the key to preserving GA's linear structure.

6. The contact-line law

Treat the contact-line radius r_c (equivalently θ_c) as a genuine dynamical variable. Its motion dissipates energy through microscopic and viscous processes near the moving line whose net effect, at leading order and for small capillary number, is a force linear in the line speed (Blake 2006; de Gennes 1985; Cox 1986; Voinov 1976); we encode this by a Rayleigh dissipation functional quadratic in the line speed,

$$\mathcal{R} = \frac{1}{2} \xi (2\pi r_c) \dot{r}_c^2 \geq 0, \quad (6.1)$$

with $\xi \geq 0$ the contact-line friction coefficient (units of $\sigma t_\sigma/R$ in dimensionless form; $\xi \rightarrow 0$ = perfectly mobile, $\xi \rightarrow \infty$ = pinned). The Lagrange–Rayleigh equation for the massless coordinate r_c is $\partial\mathcal{R}/\partial\dot{r}_c = -\partial E/\partial r_c$. We use the geometric identity

$$\frac{\partial A_{lg}}{\partial A_w} = +\cos\theta_d, \quad (6.2)$$

i.e. advancing the wetted area by dA_w changes the liquid–gas area by $+\cos\theta_d dA_w$; the sign is fixed by requiring that $\partial E/\partial A_w = 0$ at $\theta_d = \theta_e$ reproduce Young's relation (checks: complete wetting $\theta_d \rightarrow 0$ gives $+1$; perfect non-wetting $\theta_d \rightarrow \pi$ gives -1). With $A_w = \pi r_c^2$ and (4.2),

$$\frac{\partial E}{\partial r_c} = \frac{\partial}{\partial r_c} (A_{lg} - \cos\theta_e \pi r_c^2) = 2\pi r_c (\cos\theta_d - \cos\theta_e), \quad (6.3)$$

so that $\partial\mathcal{R}/\partial\dot{r}_c = \xi 2\pi r_c \dot{r}_c = -\partial E/\partial r_c = 2\pi r_c(\cos\theta_e - \cos\theta_d)$, giving the linear contact-line (Blake/de Gennes) law

$$\boxed{\xi \dot{r}_c = \cos\theta_e - \cos\theta_d.} \quad (6.4)$$

The right-hand side is the uncompensated Young force per unit length. Equation (6.4) replaces the tangency condition (2.10) as the closure for the contact line.

COROLLARY 1 (QUASI-STATIC LIMIT). *As $\xi \rightarrow 0$, (6.4) forces $\cos\theta_d = \cos\theta_e$, i.e. $\theta_d = \theta_e$: a perfectly mobile line instantaneously adopts the equilibrium angle.*

COROLLARY 2 (NON-WETTING LIMIT). *Setting $\theta_e = \pi$ in the quasi-static limit gives $\theta_d = \pi$, which by Proposition 1 is exactly the GA tangency condition (2.10). Hence ($\theta_e \rightarrow \pi$, $\xi \rightarrow 0$) recovers GA.*

COROLLARY 3 (PINNED LIMIT). *As $\xi \rightarrow \infty$, (6.4) gives $\dot{r}_c \rightarrow 0$: the contact line is pinned.*

6.1. Advancing, receding, and contact-angle hysteresis

As written, (6.4) uses a single friction ξ and a single equilibrium angle θ_e , so it is odd under $\dot{r}_c \rightarrow -\dot{r}_c$ about θ_e : the *advancing* contact line of the spreading phase and the *receding* line of the retraction phase obey the *same* relation, and the model has no contact-angle hysteresis. Real substrates instead exhibit distinct advancing and receding angles $\theta_a \geq \theta_r$ with a pinned band in between, an important and often dominant source of dissipation and of the spreading/retraction asymmetry seen in rebound (Joanny & de Gennes 1984; Snoeijer & Andreotti 2013). The present framework accommodates this at no structural cost by promoting (6.4) to a piecewise (direction- and angle-dependent) law,

$$\xi \dot{r}_c = \begin{cases} \cos\theta_a - \cos\theta_d, & \theta_d < \theta_a \quad (\text{advancing, } \dot{r}_c > 0), \\ 0, & \theta_r \leq \theta_d \leq \theta_a \quad (\text{pinned}), \\ \cos\theta_r - \cos\theta_d, & \theta_d > \theta_r \quad (\text{receding, } \dot{r}_c < 0), \end{cases} \quad (6.5)$$

optionally with distinct ξ_a, ξ_r . Equation (6.4) is the zero-hysteresis case $\theta_a = \theta_r = \theta_e$. The single-angle law is retained as the baseline here because it isolates the leading-order effect of wettability with one fitted parameter; (6.5) is the natural refinement once advancing and receding angles are measured independently, and both branches are represented in the same discrete selection residual (§8) by the sign of the contact-line increment.

7. Energy budget

PROPOSITION 4 (DISSIPATION BALANCE). *Along solutions of (2.2)–(2.5) closed by (6.4),*

$$\frac{dE}{dt} = - \underbrace{\sum_{l \geq 2} 2(2l+1)(l-1) Oh \mathcal{U}_l^2}_{\text{bulk viscous} \geq 0} - \underbrace{\xi (2\pi r_c) \dot{r}_c^2}_{\text{contact line} \geq 0} \leq 0, \quad (7.1)$$

with E from (4.1). In the inviscid, perfectly-mobile, non-wetting limit ($Oh \rightarrow 0$, $\xi \rightarrow 0$, $\theta_e \rightarrow \pi$) energy is conserved, reproducing the GA inertio-capillary limit.

Proof. The pressure does no net work on the frictionless unilateral constraint, leaving the bulk viscous sink; the wetting and moving-line liquid-gas terms combine, via the

mobility law (6.4), into the contact-line sink. The full calculation is given in Appendix A. $\square$

Equation (7.1) is the arbiter used to validate the implementation: it must close numerically, and it certifies that no spurious force (e.g. a vertical Young term in (2.5)) has been introduced — consistent with Proposition 3.

8. Discrete formulation

We inherit GA Appendix B verbatim for the bulk: truncation at mode L , implicit-Euler (or BDF) time stepping, the angular mesh $\{\theta_0 = 0\} \cup \{\text{zeros of } P_L\}$, and the block linear system $\mathbf{M}^{k+1,q} \mathbf{x}^{k+1,q} = \mathbf{b}^{k+1,q}$ for a candidate number q of contact mesh points (GA eqns for $\mathbf{M}, \mathbf{x}, \mathbf{b}$). *The extension leaves $\mathbf{M}$ and $\mathbf{b}$ unchanged.* Only the selection of q and, dynamically, its allowed rate of change are modified.

8.1. Quasi-static finite angle (perfectly mobile, $\xi = 0$)

GA select q by minimising the tangency residual (their e_q^2 , the discrete $\partial_\theta(\zeta \cos \theta)$ across the contact edge, i.e. $\propto -z_\theta$). For a finite target angle θ_e the condition $\theta_d = \theta_e$ is, by (3.2), that the tangent (x_θ, z_θ) be parallel to $(\cos(\pi - \theta_e), \sin(\pi - \theta_e)) = (-\cos \theta_e, \sin \theta_e)$; taking the (vanishing) cross product gives the *linear, well-conditioned* residual

$$e_q^2(\theta_e) = |x_\theta \sin \theta_e + z_\theta \cos \theta_e| \Big|_{\theta=(\theta_q+\theta_{q+1})/2}, \quad (8.1)$$

with x_θ, z_θ from (3.1) evaluated on the solved amplitudes $A_l^{k+1,q}$. This form is preferable to $|\theta_d^q - \theta_e|$ on three counts: it is *linear* in the amplitudes (for fixed θ_q), it is *bounded* (no division by x_θ , so it survives the $x_\theta \rightarrow 0$ spreading regime of Remark 1), and at $\theta_e = \pi$ it collapses *term for term* to GA's tangency residual: $\sin \theta_e = 0$, $\cos \theta_e = -1$ give $e_q^2 = |z_\theta| = |\partial_\theta(\zeta \cos \theta)|$, the discrete finite difference GA already use. Impenetrability e_q^1 (GA) is retained unchanged; $e_q = \max\{e_q^1, e_q^2\}$ and $m^{k+1} = \operatorname{argmin}_q e_q$.

Thus for $\theta_e = \pi$ the scheme shares GA's tangency root and reduces to GA's discrete objective; for finite θ_e only the scalar selection residual changes. No matrix $\mathbf{M}$ or right-hand-side $\mathbf{b}$ change is required, so the linear solve — and its conditioning — is that of GA, subject to the edge-load caveat of Remark 2 at angles far from π .

8.2. Dynamic contact line (finite ξ)

A finite friction must be represented *sub-mesh*: on the fixed angular grid the contact count q is an integer, so the naive choice of scoring each q by the discrete mobility imbalance pins the line for any $\xi > 0$ (a one-node change δr_c makes $\xi \delta r_c$ dominate the $\mathcal{O}(\delta_t)$ driving term). We instead carry the contact-line radius r_c as a *continuous* tracker advanced each step by a rate-limited integration of (6.4) toward the quasi-static target r_c^{qs} (the edge radius selected by (8.1)):

$$r_c^{k+1} = r_c^k + \operatorname{clamp}\left(\frac{\delta^k}{\xi} (\cos \theta_e - \cos \theta_d^k), 0, r_c^{\text{qs}} - r_c^k\right), \quad (8.2)$$

with θ_d^k the apparent angle at the current line; the integer contact set for the bulk solve is then the mesh bracket of r_c^{k+1} . The clamp gives the exact limits: $\xi \rightarrow 0$ reaches r_c^{qs} in one step (quasi-static, §8.1); $\xi \rightarrow \infty$ leaves r_c unchanged (pinned). The bulk solve at fixed bracket remains the GA linear system — the only new work is the scalar tracker update, so the well-conditioned inner solve responsible for GA's robustness is untouched, and no energy is injected into the ballistic centre-of-mass motion (verified below via the coefficient of restitution).

Optional monolithic form. Alternatively θ_c may be carried as a $(3L+2)$ -th unknown with (6.4) as an added residual row, solved by Newton with the analytic bulk Jacobian augmented by one finite-differenced column. The operator-split tracker above is the default because it cannot destabilise the proven linear core and, being sub-mesh, already resolves the continuous contact-line motion the monolithic form would provide.

9. Numerical validation

We verify the extended solver against its own limits and check the predicted trends. Except where noted, results use $L = 20$ modes, $Oh = 0.1$, $Bo = 0.01$ and impact speed $We^{1/2} = 0.5$.

(i) *Exact reduction.* With $(\theta_e = \pi, \xi = 0)$ the extended solver reproduces the GA trajectory to machine precision (the selection residual (8.1) collapses term-for-term to GA's, Appendix B), confirmed as a regression test.

(ii) *Wettability lowers restitution.* Decreasing θ_e from π at $\xi = 0$ monotonically decreases the coefficient of restitution and increases the contact time and maximal contact radius:

θ_e/π	1.00	0.96	0.92	0.90
COR	0.657	0.654	0.552	0.361
t_c	2.60	2.63	2.96	3.40
β_c	0.516	0.516	0.654	0.683

consistent with the loss of rebound energy to increased spreading as the surface becomes more wettable.

(iii) *Friction resists capture.* At fixed $\theta_e = 0.9\pi$, increasing the contact-line friction ξ monotonically increases the coefficient of restitution, from the quasi-static value (0.361 at $\xi = 0$) up to the perfectly non-wetting value (0.657) once the line is effectively pinned ($\xi \gtrsim 1$); the maximal contact radius correspondingly falls back to its non-wetting value. This realises the $\xi \rightarrow 0$ (quasi-static) and $\xi \rightarrow \infty$ (pinned) limits of (8.2) and shows contact-line friction acting as a distinct, tunable dissipation channel.

(iv) *No energy injection.* In every case above the coefficient of restitution satisfies $COR \leq 1$, the operational form of the energy budget (Prop. 4): no spurious energy enters the ballistic centre-of-mass motion, including at the finite contact angles where the edge-load truncation error (5.1) is largest. Quantitative accuracy is expected to be best near $\theta_e = \pi$ (error $\mathcal{O}(\sin \theta_d)$, Remark 2); the trustworthy band for the parameters here is $\theta_e \gtrsim 0.9\pi$.

10. Summary of the extended model

The extended KM model is (2.2)–(2.5) together with the contact constraints (2.7)–(2.9) and the contact-line law (6.4) in place of the tangency condition (2.10), with θ_d given by (3.2). It has:

- one new dynamical degree of freedom, the contact-line radius r_c ;
- two physical parameters, the equilibrium angle θ_e (directly measurable) and the contact-line friction ξ (a single fit parameter);
- an exact reduction to GA at $(\theta_e \rightarrow \pi, \xi \rightarrow 0)$ (Cor. 2);
- a COM momentum equation that retains the GA form, the contact-line reaction being absorbed into $\mathcal{B}_1$ (Prop. 3);
- a closed energy budget with a distinct contact-line dissipation channel (Prop. 4);
- a discrete form that changes only the contact-set selection criterion (§8), preserving GA's linear solve.

Appendix A. Full derivation of the energy budget

We give the calculation sketched in Proposition 4. Differentiate the kinetic and capillary contributions of (4.1). Using the modal energy $e_l = \frac{1}{2}\mathcal{U}_l^2 + \frac{1}{2}l(l+2)(l-1)\mathcal{A}_l^2$ and the equations of motion (2.2)–(2.3),

$$\dot{e}_l = \mathcal{U}_l \dot{\mathcal{U}}_l + l(l+2)(l-1)\mathcal{A}_l \dot{\mathcal{A}}_l = -2(2l+1)(l-1)Oh \mathcal{U}_l^2 - l\mathcal{B}_l \mathcal{U}_l, \quad (\text{A } 1)$$

so summing over $l \geq 2$ and adding $\dot{E}_{\text{grav}} + \frac{1}{2} \frac{d}{dt} v^2 = Bo v + v\dot{v} = v\mathcal{B}_1$ (from (2.5)) gives

$$\dot{E}_{\text{bulk}} = - \sum_{l \geq 2} 2(2l+1)(l-1)Oh \mathcal{U}_l^2 - \left(\sum_{l \geq 2} l\mathcal{B}_l \mathcal{U}_l - v\mathcal{B}_1 \right). \quad (\text{A } 2)$$

The parenthesised term is the rate of working of the contact pressure on the interface. On the contact set the constraint fixes the normal position ($z = 0$), so the interface normal velocity vanishes wherever $p \neq 0$; off the contact set $p = 0$. Hence $\int p u_n dS = 0$ and the pressure-work term vanishes identically (frictionless unilateral contact does no net work), leaving only the bulk viscous sink. Finally the surface-wetting energy: from (4.2) and (6.2),

$$\dot{E}_{\text{surf}} = \underbrace{\frac{\partial A_{lg}}{\partial r_c} \dot{r}_c}_{+2\pi r_c \cos \theta_d \dot{r}_c} - \cos \theta_e 2\pi r_c \dot{r}_c = -2\pi r_c (\cos \theta_e - \cos \theta_d) \dot{r}_c = -\xi (2\pi r_c) \dot{r}_c^2, \quad (\text{A } 3)$$

the last equality by the mobility law (6.4). Adding the two contributions yields (7.1). Both sinks are manifestly non-negative, and each vanishes in its respective limit ($Oh \rightarrow 0$; $\xi \rightarrow 0$ or $\dot{r}_c \rightarrow 0$).

Appendix B. The discrete selection residual and its reduction to GA

GA (their Appendix B) solve, for each candidate contact count q , the linear system $\mathbf{M}^{k+1,q} \mathbf{x}^{k+1,q} = \mathbf{b}^{k+1,q}$ and select q by minimising the tangency residual $e_q^2 = |Z(\theta_q) - Z(\theta_{q+1})|$, with $Z(\theta) = \zeta(\theta) \cos \theta + z$ the collocated height. Our extension leaves $\mathbf{M}, \mathbf{b}$ untouched and replaces only e_q^2 . Writing the discrete edge differences $\Delta Z = Z(\theta_q) - Z(\theta_{q+1})$ and $\Delta x = x(\theta_q) - x(\theta_{q+1})$ with $x(\theta) = \zeta(\theta) \sin \theta$, the finite-angle residual is

$$e_q^2(\theta_e) = |\Delta x \sin \theta_e + \Delta Z \cos \theta_e|, \quad (\text{B } 1)$$

the discrete form of (8.1). Two properties follow directly. (i) *Exact GA reduction*: at $\theta_e = \pi$, $\sin \theta_e = 0$ and $\cos \theta_e = -1$, so $e_q^2(\pi) = |\Delta Z|$, GA's tangency residual term for term; the selected q , and hence the entire trajectory, are identical. (ii) *Boundedness*: (B 1) is a linear combination of collocated amplitude differences, with no division by x_θ , so it remains well defined through the maximal-spreading regime where $x_\theta \rightarrow 0$ (Remark 1). For the hysteretic law (6.5), the same residual is used with θ_e replaced by θ_a or θ_r according to the sign of the trial contact-line increment $r_c^{k+1,q} - r_c^k$, and pinned candidates ($\theta_r \leq \theta_d \leq \theta_a$) are scored by their distance to the nearest branch.

REFERENCES

- Alventosa, L.F.L., Cimpeanu, R. & Harris, D.M. 2023 Inertio-capillary rebound of a droplet impacting a fluid bath. *J. Fluid Mech.* **958**, A24.
 Blake, T.D. 2006 The physics of moving wetting lines. *J. Colloid Interface Sci.* **299**, 1–13.
 Cox, R.G. 1986 The dynamics of the spreading of liquids on a solid surface. Part 1. Viscous flow. *J. Fluid Mech.* **168**, 169–194.

- Gabbard, C.T., Agüero, E.A., Cimpeanu, R., Kuehr, K., Silver, E., Barotta, J.-W., Galeano-Rios, C.A. & Harris, D.M. 2025 Drop rebound at low Weber number. *J. Fluid Mech.* (companion paper).
- Galeano-Rios, C.A., Milewski, P.A. & Vanden-Broeck, J.-M. 2017 Non-wetting impact of a sphere onto a bath and its application to bouncing droplets. *J. Fluid Mech.* **826**, 97–127.
- de Gennes, P.-G. 1985 Wetting: statics and dynamics. *Rev. Mod. Phys.* **57**, 827–863.
- Grmela, M. & Öttinger, H.C. 1997 Dynamics and thermodynamics of complex fluids. I. Development of a general formalism. *Phys. Rev. E* **56**, 6620–6632.
- Joanny, J.F. & de Gennes, P.-G. 1984 A model for contact angle hysteresis. *J. Chem. Phys.* **81**, 552–562.
- Lamb, H. 1932 *Hydrodynamics*, 6th edn. Cambridge University Press.
- Moláček, J. & Bush, J.W.M. 2013 Drops bouncing on a vibrating bath. *J. Fluid Mech.* **727**, 582–611.
- Okumura, K., Chevy, F., Richard, D., Quéré, D. & Clanet, C. 2003 Water spring: a model for bouncing drops. *Europhys. Lett.* **62**, 237–243.
- Richard, D., Clanet, C. & Quéré, D. 2002 Contact time of a bouncing drop. *Nature* **417**, 811.
- Snoeijer, J.H. & Andreotti, B. 2013 Moving contact lines: scales, regimes, and dynamical transitions. *Annu. Rev. Fluid Mech.* **45**, 269–292.
- Voinov, O.V. 1976 Hydrodynamics of wetting. *Fluid Dyn.* **11**, 714–721.